

Deep Learning-Based Identification of SSB-PriA Interaction Inhibitors Against Multidrug-Resistant *Klebsiella pneumoniae* Using Graph Neural Networks

Ruchi Shah (Lead)¹, Isha Shah¹, Krutik Babariya¹, Dilkhush Yadav¹

¹Faculty of Applied Sciences & Technology, Humber Polytechnic, Toronto, Canada

{n10020731, n10009347, n10009322, n01723579}@humber.ca

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Abstract - The rate at which antibiotic resistance is increasing surpasses the rate of new drug development, and *Klebsiella pneumoniae* is at the centre of this global epidemic. The bacterium is a gram-negative resistant to practically all classes of antibiotics, including carbapenems, the last-resort choice, declared by the World Health Organization a critical-priority pathogen [1]. The project is focused on the SSB-PriA protein-protein interaction (PPI) which is required to restart bacterial DNA replication and is missing in human cells but is preserved in gram-negative pathogens [2]. One hybrid deep learning model (FinalUltraModel) was trained on PubChem BioAssay AID 1272365 that contained about 440,000 compounds experimentally tested against SSB-PriA. The model combines Graph Attention Networks (GAT) of molecular graphs encoding and Morgan fingerprint vectors of global chemical representation. A 1:2 active-inactive balanced subset of 7,785 molecules was made to deal with serious class imbalance (0.78% actives). The model was trained with the Adam optimizer and 80/20 split, reaching an accuracy of 86.06, AUC-ROC of 0.8800, Precision of 0.8102, Recall of 0.7216, and F1-score 0.7634, indicating a strong discriminative property in virtual screening without laboratory experimentation.

Index Terms - *Klebsiella pneumoniae*; Antimicrobial Resistance; SSB-PriA Protein-Protein Interaction; Graph Neural Networks (GNN); Graph Attention Network (GAT); Molecular Graph Representation; Drug Discovery; Virtual Screening; Deep Learning; Morgan Fingerprints; AID 1272365; PubChem BioAssay; RDKit; Class Imbalance; Computational Pharmacology.

I. INTRODUCTION

A. Antimicrobial Resistance.

One of the most urgent public health crises in the 21st century is antibiotic resistance. Think of the scenario whereby a patient is admitted in a hospital and the physicians start using all the possible effective antibiotics one after another- that is no longer a hypothetical scenario but a growing reality. *Klebsiella pneumoniae* is a gram-

negative bacterium that is commonly present in healthcare facilities and causes serious infections, such as pneumonia, bloodstream infections, and urinary tract infections. Carbapenem-resistant strains have a mortality rate of up to 50% [1] and the World Health Organization (WHO) has put these strains on its critical-priority list. The traditional method of antibiotic development, which has been mainly the derivatives of already existing drugs, is not able to match the adaptation of bacteria, and so the really new targets are essential.

B. The SSB-PriA Interaction as a New Antibiotic Target.

The PriA helicase-Single-stranded DNA-binding protein (SSB) interaction was found to be an attractive antibiotic target in *K. pneumoniae*. SSB shields exposed DNA strands in the context of replication; PriA is a helicase that resumes stalled replication forks in case of bacterial stress. This interaction is interrupted and inhibits the repair and replication of DNA causing bacterial death [2]. Most importantly, the target is biologically critical, preserved in gram-negative pathogens, and not found in a similar form in human cells- reducing harm to the patient. Although the interface with high therapeutic potential, the SSB-PriA interface has not been previously tackled with any computational deep learning pipeline in virtual screening.

C. Virtual Screening with a Deep Learning Solution.

It would take years and colossal resources to physically test 440,000 compounds in a laboratory. Alternatively, a Graph Neural Network (GNN) model was trained to predict molecular activity with solely chemical structure as input, such that a computational reduction of candidates can be performed prior to laboratory work commencing. GNNs model molecules as graphs (atoms are nodes, bonds are edges) and directly learn structural motifs. Graph Attention Networks (GAT) [3] take this further by having learned attention coefficients, allowing focused attention to be given to pharmacophoric substructures. Together with Morgan fingerprints encoding global chemical topology, FinalUltraModel offers two complimentary molecular insights into each candidate compound.

D. Industry Partnership

This is a joint-production with Living In Silico Inc. a computational drug discovery company. They aimed to create a model that is trained on actual experimental high-throughput screening (HTS) data that would be able to predict SSB-PriA inhibitors with reasonable accuracy, allowing the researchers to divert laboratory effort to the computationally identified hits most likely to succeed- especially in early-stage drug discovery where the chemical space is large and limits on resources are high.

II. LITERATURE REVIEW

A. Overview of Field.

The last ten years have been associated with a great transformation in computational drug discovery due to the increased availability of large biological data and developments in machine learning [4]. The conventional pipeline, high-throughput screening (HTS), identification of hits, optimization of leads, is still expensive, slow and fails to keep up with bacterial evolution [14]. Machine learning-based virtual screening is a more scalable and faster alternative: instead of experimentally testing every compound, the models are trained on available bioassay measurements and applied to predict the activity of novel compounds prior to starting laboratory work.

Graph Neural Networks (GNNs) have been found especially effective in the prediction of molecular properties. GNNs, in contrast to SMILES strings or fixed-length descriptor vectors, incorporate structural and relational information that has been lost in other representations, and model molecules as atom and bond networks. Duvenaud et al. [8] showed that graph convolutional networks on molecular graphs can be as good or even better on standard circular fingerprints than on benchmark problems. Gilmer et al. [5] combined these methods into the general Message Passing Neural Network (MPNN) framework. Xu et al. [9] established theoretical limits on expressive power of GNN and presented the Graph Isomorphism Network (GIN).

Nevertheless, these initial investigations employed small, clean and balanced datasets- conditions that are not representative of large-scale pharmaceutical HTS data.

The Graph Attention Network (GAT) by Veličkovic et al. [3] presented learned attention coefficients that allow the differentiation of contributions of neighbours when passing the message. In molecular graphs, this enables the model to specifically target the most chemically important substructures of atoms- which is especially useful when pharmacophoric features of interest in PPI inhibition only occupy a subset of the molecular graph. This was one of the reasons why we decided to employ GAT as the core encoder in FinalUltraModel.

The Stokes et al. [10] work is a breakthrough validation of deep learning in antimicrobial discovery. A directed MPNN trained on the Drug Repurposing Hub identified halicin, a structurally novel compound with broad-spectrum bactericidal activity that was later tested in animals. Nevertheless, the latter model focused on whole-cell phenotypic inhibition with a moderate-sized and relatively small training set. Filtering a large, grossly imbalanced HTS screen of inhibitors of a particular protein-protein interaction, such as in this project, has significantly more difficult technical demands.

Hwang et al. [7] compared the GNN architectures systematically on the tasks of molecular learning and found that the AUC-ROC and F1-score are more reliable measures of true classifier quality when the actives are a small fraction of the original AID 1272365 data-point set- indeed, just that- and indeed, precisely that. Kearnes et al. [12] also demonstrated that a graph-based representation coupled with molecular fingerprints is always superior to either representation and thus the decision to use GAT embeddings in conjunction with Morgan fingerprints.

B. Critical Analysis of Significant Literature.

Table I reports a systematic critical review of the major references, which directly informed FinalUltraModel design decisions.

Reference	Problem & Objective	Method & Data	Key Results	Critical Limitations	Applicability
Duvenaud et al. (2015) [8]	Graph-based fingerprints vs hand-crafted descriptors for bioactivity prediction.	GCN; small ADMET benchmark data.	Surpassed circular fingerprint performance on several tasks.	Small, balanced data; no class imbalance treatment; no PPI targets; no HTS-scale validation.	Provides foundational GCN concepts. This paper scales graph learning to large-scale PPI screening with imbalance mitigation.
Gilmer et al. (2017) [5]	Standardize GNN designs	QM9 quantum chemistry database	State-of-the-art on 11 of 13 quantum chemistry	Targets continuous quantum chemistry	Message passing forms the theoretical basis of the GAT layers in FinalUltraModel,

Reference	Problem & Objective	Method & Data	Key Results	Critical Limitations	Applicability
	into the MPNN framework.	(~130K molecules).	regression objectives.	properties on balanced data; not binary bioactivity with high class imbalance.	adapted here for imbalanced binary classification.
Xu et al. (2019) [9]	Set theoretical power bounds of GNNs; propose GIN as most discriminative architecture.	GIN; bioinformatics and social network benchmarks.	GIN outperforms GCN and GraphSAGE on graph classification.	Small, class-balanced benchmarks; no evaluation on pharmaceutical HTS data or PPI targets.	Informs model selection trade-offs; GAT chosen over GIN for its selective attention on molecular substructures.
Stokes et al. (2020) [10]	Discover structurally novel antibiotics via deep learning virtual screening.	MPNN; Drug Repurposing Hub (~2,500 compounds).	Halicin—a novel broad-spectrum antibiotic—discovered and validated in animals.	Small, moderately balanced dataset; targets whole-cell phenotypic inhibition, not a specific PPI interface.	Validates DL-based antibiotic discovery. This work scales to a larger, precisely experimental PPI screening dataset.
Hwang et al. (2020) [7]	Compare GNN architectures systematically on supervised molecular property prediction.	GCN, GAT, MPNN variants; MoleculeNet benchmark suite.	GAT competitive; AUC-ROC and F1 more faithful than accuracy in imbalanced settings.	Benchmark data with 5–30% active rates—less extreme than AID 1272365's 0.78%. No hybrid models or PPI targets.	Directly informs evaluation strategy: AUC-ROC and F1 selected as primary metrics for imbalanced bioactivity data.
Kearnes et al. (2016) [12]	Demonstrate superiority of hybrid graph-fingerprint representations.	Molecular graph convolutions + fingerprints on standard ADMET benchmarks.	Hybrid representations consistently outperform either alone.	Smaller, class-balanced benchmark datasets; not applied to large-scale HTS imbalanced PPI data.	Directly motivates FinalUltraModel's dual-branch design combining GAT embeddings with Morgan fingerprints.

TABLE I: Critical Analysis of Key Literature Supporting FinalUltraModel Design Decisions

C. Research Gaps Found.

Gap 1 : No SSB-PriA-specific deep-learning: To the best of the existing knowledge, no previous research has used Graph Neural Networks (GNNs) to the virtual screening of SSB-PriA inhibitors with experimental high-throughput screening (HTS) data. The dataset AID 1272365 has never been investigated with the help of the graph-based deep learning strategies.

Gap 2 : Extreme imbalance in classes at HTS scale: The majority of the studies of GNN-based learning of molecules report active compound rates in 5-30%. The AID 1272365

dataset, by contrast, has an active rate of 0.78% which is a very imbalanced real-world HTS situation. Limited literature has been done on this degree of imbalance on protein-protein interaction (PPI) targets, especially in gram-negative pathogens.

Gap 3 : Limited hybrid GAT architectures: Hybrid graphfingerprint representations have shown better performance, but applied on multi-layer Graph Attention Network (GAT) structures with global fingerprint features on large-scale, imbalanced PPI bioassay datasets is understudied.

Gap 4 : SSB-PriA as a computational target: Although the SSB-PriA interaction is a highly biological target due to its

conservation and necessity in gram-negative pathogens [2], it has received little computational drug discovery coverage. The present project will directly fill this gap by creating a specific deep learning pipeline to predict SSB-PriA inhibitors.

III. Problem statement and motivation of the research.

Klebsiella pneumoniae with multidrug resistance is a dire and growing health crisis across the world. As the resistance to last-resort carbapenem antibiotics is becoming global, and the traditional drug development cannot keep up with the bacterial evolution, the new antibiotic targets and effective ways to find new candidate inhibitors are urgently needed. One such target is the SSB-PriA protein-protein interaction (PPI): this interaction is necessary to restart bacterial DNA replication, is not functionally recapitulated in human cells, and is conserved among gram-negative pathogens. Nonetheless, although promising as a therapy, no computational deep learning pipeline has hitherto focused on this interaction in virtual screening.

The major technical difficulty is the form of experimental data available. The PubChem BioAssay AID 1272365 database includes about 440,000 compounds tested on SSB-PriA under laboratory conditions, yet only about 0.78 percent are active, which is a drastic imbalance in the classes, rendering the typical machine learning algorithms unreliable. The usual accuracy measures are invalid in a situation where a naive classifier can obtain high accuracy by classifying all compounds as inactive.

Thus, the overarching issue that the study aims to address is: How can a deep learning model be developed, trained, and validated to effectively select SSB-PriA interaction inhibitors out of a massively imbalanced high-throughput screening dataset, and provide a reliable computational resource to drug discovery in the early stages of dealing with multidrug-resistant *K. pneumoniae*?

IV. Research questions and objectives.

The main objectives and research questions that this project will respond to are as follows:

Objective 1: Prepare FinalUltraModel- a deep-learning model that is a hybrid of Graph Attention Networks and Morgan fingerprint processing in binary SSB-PriA inhibitor classification.

Objective 2: Create and run a preprocessing and class-balancing pipeline that is able to address the extreme class imbalance (0.78 active rate) occurring in the AID 1272365 dataset.

Objective 3: Train and test the model with the help of proper metrics (AUC-ROC, F1-score, Precision, Recall)

which have been verified by Hwang et al. [7] to work with imbalanced bioassay data.

Objective 4: Attain an AUC-ROC of over 0.80 and accuracy of over 85 that would reflect a significant discriminative capability of an active and inactive compound as a useful virtual screening system.

Research Question 1: Does a hybrid GAT-fingerprint model that has been trained on balanced HTS data reliably discriminate between SSB-PriA inhibitors and non-inhibitors?

Research Question 2: What is the trade-off between precision and recall in a severely imbalanced drug discovery setting and what does this mean to practical utility to Living In Silico Inc?

V. MATERIALS and methods.

This project follows a systematic deep learning research procedure to discover possible inhibitors of the SSB-PriA protein-protein interaction in multidrug-resistant *Klebsiella pneumoniae*. The methodology includes data collection and preprocessing, building of molecular graphs, hybrid features extraction, training of the model with mitigation of class-imbalance, validation, and evaluation.

A. Data Acquisition and Preprocessing.

The PubChem BioAssay AID 1272365 dataset [15] was downloaded, which has experimental screening data of about 440,000 compounds that have been tested in the SSB-PriA interaction. The SMILES strings were converted to graph representations using RDKit, with atoms and chemical bonds represented as nodes and edges respectively. The atom-level features that have been encoded are: the type of atom (one-hot encoding of C, N, O, S, F, Cl), the degree, and the aromaticity, resulting in 10-dimensional node feature vectors. This representation of features increases the ability of the model to comprehend chemical structure and biological relevance. A total of 441,244 records of compounds were parsed after SMILES validation and preprocessing, retaining valid records containing SMILES and activity labels.

B. Mitigation of Class Imbalance.

The initial dataset had drastic imbalance of classes. This was dealt with by undersampling randomly inactive compounds with a fixed random seed (42) to reproducibly achieve this. A subset of 7,785 molecules (2,595 active and 5,190 inactive compounds) was then formed by selecting an equal number of active and inactive molecules in a balanced set with an active to inactive ratio of about 1:2. This ratio was chosen empirically to have meaningful negative examples, but not to allow the majority class to subsume the model, and to avoid overtraining the model due to having too many chemical examples.

C. Molecular Feature Representation.

Each compound was represented using two complementary molecular representations:

Graph Representation: The molecules were transformed into a PyTorch Geometric Data object, consisting of node feature tensors (10-dimensional atom features), edge index tensors that encode bond connectivity (bidirectional), binary activity labels, and corresponding Morgan fingerprint vectors.

Morgan Fingerprint Vectors: 2048-bit Morgan fingerprint (radius = 2) were calculated with RDKit AllChem module, encoding the global chemical substructure around each atom. These offer a complement to graph learning a local structural fingerprint, at the molecular level, on a chemical fingerprint.

D. Dataset Splitting

The homogeneous data set of 7,785 molecules was divided into a training and a testing sample with a 80/20 ratio, to obtain 6,228 training samples and 1,557 testing samples. The division was done with shuffling a fixed random seed (42) to make them reproducible. A batch size of 16 was applied to a DataLoader to effectively batch graphs through PyTorch Geometric.

E. Training Configuration

Training of the model was made possible by Adam optimizer (learning rate= 0.00003, weight decay= 1e-4). The maximum number of epochs trained was 50 and the desired accuracy threshold was 86. The model was stored in case the target accuracy was achieved. Cross-Entropy Loss was used as training objective. The seed was kept constant (42) to ensure reproducibility. CPU hardware training was done with RDKit warnings silence so that the output would be cleaner.

VI. FRAMEWORK AND ARCHITECTURE

FinalUltraModel is a graph based structural learning, fingerprint based chemical representation hybrid molecular classification framework. The architecture has two parallel branches the output of which is concatenated and fed to a common classification network. PyTorch implementation of the model is done using PyTorch Geometric.

A. Graph Attention Network (GAT) Branch

The former branch uses two sequential Graph Attention Network (GATConv) layers to treat molecular graphs. The message-passing of each layer is to the neighboring atoms and learned attention coefficients are to show the relative contribution of the features of the neighbor atoms. The particular architecture is:

- GATConv(10 64, 4 attention heads)+ ELU activation 256-dim out.
- GATConv(256 128, 1 attention head) ELU activation - 128-dim output.

After the last GAT layer, global mean pooling is used to generate a 128-dimensional graph embedding, which encodes the overall structural characteristics of the whole molecular graph.

B. Morgan Fingerprint Branch

The second branch compares 2048-bit Morgan fingerprint vectors with a fully connected neural network: 2048 512 dimensions, which uses ReLU activation and Dropout (0.5). There is global substructure of chemical patterns that this branch captures but which are not explicitly local graph topology.

C. Joint Classification Network.

The 128-dimensional graph embedding and the 512-dimensional fingerprint embedding are then concatenated to make a 640-dimensional combined representation. It is run through a sequence of denser layers (640 to 256 to 2) where the activations are ReLU and the Dropout is set to 0.6 and finally classifies a binary (Active/Inactive). The hybrid architecture allows the model to jointly use local structural features of the graph learning process and global chemical features of the fingerprint descriptors.

VII. DATASET DESCRIPTION

The dataset in this project is the PubChem BioAssay AID 1272365 dataset which is a summary of experimental screening data of compounds against SSB-PriA protein-protein interaction.

A. Source and Origin.

Host database AID 1272365 can be found in the PubChem database of the National Center of Biotechnology Information (NCBI) [15]. It is an actual high-throughput screening program whereby compounds were experimentally tested to inhibit the SSB-PriA interaction under laboratory conditions.

B. Data Characteristics

The compounds are encoded by their SMILES strings that encode the chemical structure and a binary activity label (Active = 1 / Inactive = 0). The initial dataset has about 441, 244 entries of compounds under 12 columns such as PUBCHEM_SID, PUBCHEM_CID, SMILES, activity outcome, concentration data, and percent inhibition replicates. In the preprocessing stage, metadata rows were eliminated and records that lacked SMILES strings or activity labels were filtered out to maintain quality data to use in training the models.

C. Balanced Subset to be used to train.

With such a high degree of class imbalance, a balanced subset of 7,785 molecules was obtained by strategic undersampling. This subgroup was divided into training and testing sets 80/20. RDKit was used to compute both molecular graph representation and 2048-bit Morgan fingerprint of all compounds.

Parameter	Value
Original Dataset Entries	~441,244 compounds
Dataset Columns	12 (SMILES, activity outcome, SID, CID, concentration, etc.)
Active Compounds (original)	~0.78% of valid entries
Balanced Training Subset	7,785 (active:inactive ratio $\approx$ 1:2)
Train / Test Split (80/20)	6,228 training / 1,557 testing
Molecular Representations	10-dim node features + 2048-bit Morgan fingerprints

TABLE II: Dataset Summary — AID 1272365

D. Ethical, Privacy and Licensing.

PubChem AID 1272365 dataset is freely accessible, and has no privacy or personal information implications, since it only includes chemical structures and experimental bioassay outcomes of the chemical compounds. There was no proprietary or commercially restricted information. The data is used and accessed to conduct scholarly research as per the NCBI PubChem open access policy.

VIII. Implementation and Experimental Set Up.

FinalUltraModel was written in Python with PyTorch 2.x, PyTorch Geometric 2.7.0, RDKit 2026.3.1, NumPy 2.3.5, Pandas 2.3.3 and Scikit-learn 1.7.2. It was developed in a Jupyter Notebook (Anaconda distribution) on a Windows platform using CPU-only hardware.

A. Data Loading and Processing.

The PyTorch Geometric DataLoader with a batch size of 16 was used to load data, and it was possible to efficiently batch heterogeneous graph data. RDKit Chem was used to convert

SMILES to graph.MolFromSmiles() and AllChem.GetMorganFingerprintAsBitVect() functions. Preprocessing filtered out invalid SMILES strings. The transformation of 7,785 molecules to graph objects was done in about 58 seconds with a rate of 132 molecules per second.

B. Training Setup

The model was allowed to train up to 50 epochs but terminated early when the accuracy target of 86% was achieved. Learning rate was set to 3×10^{-5} and weight decay 1×10^{-4} . Cross-Entropy Loss was used as the training objective. The last configuration did not have any gradient clipping or learning rate scheduling, because the extremely small learning rate gave naturally stable training dynamics.

C. Training Progression

The model was very stable in its improvement in the course of training and the performance improved gradually with each epoch. There was a gradual reduction in training loss at Epoch 1 of 0.6399 to less than 0.0192 at the last Epoch of 33 which proved successful learning. During training, the test accuracy was 84-86 percent, and it reached the target threshold. The model had a test accuracy of 86.06 which is within the accuracy target of the project. The model checkpoint at this optimum performance point (best_model_8510.pth) was used to evaluate the final performance.

IX. RESULTS

The held-out test set of 1,557 molecules was tested on FinalUltraModel. The evaluation of performance was based on several different metrics that are suitable in the imbalanced binary classification task: Accuracy, Precision, Recall, F1-score, and AUC-ROC, which is compatible with the methodology that Hwang et al. [7] suggested to evaluate imbalanced bioactivity classification.

A. Performance Metrics

Metric	Value
Accuracy	86.06% (0.8606)
Precision	0.8102
Recall	0.7216
F1-Score	0.7634
AUC-ROC	0.8800

TABLE III: FinalUltraModel Performance Metrics on the Held-Out Test Set

B. Confusion Matrix

	Predicted Inactive	Predicted Active
Actual Inactive	990 (TN)	82 (FP)
Actual Active	135 (FN)	350 (TP)

TABLE IV: Confusion Matrix for FinalUltraModel on Test Set

The confusion matrix indicates that the model made the right decision on 990 inactive compounds (True Negatives) and 350 active compounds (True Positives). One hundred and thirty five active compounds were mismatched with inactive (False Negatives), and eighty two inactive compounds were mismatched with active (False Positives).

The difference between the recall (0.7216) and precision (0.8102) indicates that the model is more discriminative in predicting active compounds. This minimizes false positives and causes some true active compounds to be missed. This trade-off between precision and recall can be seen in the resulting F1-score of 0.7634.

The AUC-ROC of 0.8800 can be distinguished as a high ranking performance at all decision thresholds.

X. DISCUSSION

FinalUltraModel results indicate that it is possible to model a challenging drug discovery problem with hybrid graph-based deep learning models, especially in protein-protein interaction (PPI) inhibition screening which has a serious class imbalance.

A. Interpretation of Main Results.

The highest achievement of the model is the AUC-ROC of 0.8800. This measure indicates how well the model can put the active compounds higher than the inactive ones at all the possible classification thresholds- the most important practical need of a virtual screening tool. The AUC-ROC of 0.8800 is far more than the random performance (0.5) and the target 0.80 set in the project objectives. This finding is especially interesting in the light of the extreme level of imbalance between classes and the natural complexity of the task of predicting PPI inhibition.

The total accuracy of 86.06% also exceeds the 86% target. The model is more precise (0.8102) than recall (0.7216) thus demonstrating that it will be less aggressive in its prediction strategy. This minimizes false positives with the additional advantage of detecting a good fraction of true active compounds.

The complementary nature of the information sources integrated by the hybrid architecture can be seen as the value

of the hybrid architecture. GAT layers can identify local structural relationships in molecular graphs, allowing the model to identify chemically important substructures that are associated with SSB-PriA binding activity. The Morgan fingerprint branch also encodes global chemical topology, which can give a larger context than can be captured by individual atom neighborhoods. This combination of theirs is always more effective than single-representation methods, as Kearnes et al. [12] confirm.

B. Precision-Recall Trade-off

Precision-recall profile is skewed with precision (0.8102) larger than the recall (0.7216). Recall is also relevant in drug discovery processes since false negatives are promising drug candidates that could be overlooked. But increased precision decreases the false positives which are compounds that are not required to be put through experimental validation. This balance enhances efficiency in the screening in the downstream and still recovers a significant fraction of true active compounds.

The practical aim of the model does not involve it being perfect in classification but rather the substantial reduction of the experimental testing load. Living In Silico Inc. can expand laboratory resources on fewer and higher-quality candidates than random selection because the priorities are based on compounds that are predicted to be active.

C. Comparison with Literature.

This AUC-ROC of 0.8800 is favourable when compared to similar computational drug screening studies. Stokes et al. [10] observed good results on a smaller and less imbalanced dataset (~2,500 compounds with moderate balance in terms of classes). This work shows that a good level of discriminative performance can be obtained even with the scale of AID 1272365, and the active rate is significantly lower than most benchmark datasets. The obtained AUC-ROC is significantly above the project objective level of 0.80.

D. Practical Implications

In the case of Living In Silico Inc., FinalUltraModel can be used as a convenient first-pass screening tool. Instead of experimentally testing tens of thousands of compounds, researchers can use the model to rank compound libraries and experimentally validate the top-ranking compounds. The strategy will significantly decrease screening expenses and timelines in first-stage drug discovery programs against SSB-PriA in multidrug-resistant *K. pneumoniae*. The AUC-ROC (0.8800) of the model is used to establish that there will be a meaningfully high probability of a true biological activity of the prioritized candidates compared to random selection.

XI. LIMITATIONS

Even though the results are encouraging, there are a number of limitations affecting the performance and generalizability of this project:

Computational-Only Validation: All results are based on computational predictions. The model has not been tested in the laboratory; the active compounds predicted to occur may not be found to be actually active in wet-lab studies.

Features of simplified atoms: Node features represent atom type (one-hot with 6 element types), degree and aromaticity (10 dimensions in total). More chemically meaningful patterns could be captured in richer representations including formal charge, membership of rings, number of hydrogen and implicit valence.

Lack of 3D Structural Information: The model uses only 2D molecular graph and Morgan fingerprints. The current approach does not capture 3D molecular conformations, which are very sensitive to protein-protein interactions.

Binary Classification Limit: Simple active/inactive labels make the problem too simple. Compounds that are just above the threshold of activity can be misclassified.

Small Computational Resources: Training was done on CPU hardware, which restricted exploration of larger architectures and large-scale hyperparameter optimization.

No External Validation Set: The model was tested on an 80/20 split of the same balanced data.

Also, the model was tested on a half-screen of the sample which can lead to positive performance predictions than actual screening conditions.

XII. CONCLUSION

The gap that was filled in this project was that there was no comprehensive computational screening pipeline based on deep learning to discover SSB-PriA interaction inhibitors against multidrug-resistant *Klebsiella pneumoniae*.

FinalUltraModel was designed as a hybrid architecture that used Graph Attention Networks with a branch of Morgan fingerprint processing encoding of local structural and global chemical features. The model was trained on a balanced set of 7,785 molecules and tested on a held-out test set of 1,557 molecules. It had a Precision of 0.8102, Recall of 0.7216 and an F1-score of 0.7634 with an accuracy of 86.06, AUC-ROC of 0.8800, which is within the performance targets of the project.

The largest outcome is the model AUC-ROC score of 0.8800, as it indicates that this model can be depended upon to rank active compounds higher than inactive ones- the key objective of the virtual screening. In the case of Living In Silico Inc., this allows one to focus on high-quality candidate compounds, which will reduce the amount of

experimental work and speed up the process of early-stage drug discovery.

The work can be extended in the future to add more features of atoms, 3D molecular descriptors, experiment with more advanced graph architectures, and test predictions on independent datasets.

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Appendix A: Model Architecture Overview

Component	Details
Graph Branch – Layer 1	GATConv (10 → 64, 4 attention heads) + ELU activation
Graph Branch – Layer 2	GATConv (256 → 128, 1 attention head) + ELU activation
Global Pooling	global_mean_pool → 128-dimensional graph embedding
Fingerprint Branch	Input: 2048-bit Morgan Fingerprint → Linear(2048, 512) + ReLU + Dropout(0.5)
Feature Concatenation	128 (graph) + 512 (fingerprint) = 640-dimensional combined representation
Joint Classifier – Layer 1	Linear(640, 256) + ReLU + Dropout(0.6)
Joint Classifier – Output	Linear(256, 2) → class logits (Active / Inactive)

Appendix Table A1: FinalUltraModel Architecture Summary

Appendix B: Training Hyperparameters

Parameter	Value
Optimizer	Adam (torch.optim.Adam)
Learning Rate	3×10^{-5} (0.00003)
Weight Decay	1×10^{-4}
Batch Size	16
Max Epochs	50 (early termination at 86% accuracy threshold)
Loss Function	Cross-Entropy Loss (F.cross_entropy)
Random Seed	42 (torch.manual_seed)
Activation Functions	ELU (GAT layers), ReLU (classifier)
Dropout Rates	0.5 (fingerprint branch), 0.6 (classifier)

Appendix Table B1: Training Hyperparameters

Appendix C: Dataset Summary

Parameter	Value
Original Dataset Entries	~441,244 compounds
Dataset Columns	12
Active Compounds (balanced set)	2,595
Inactive Compounds (balanced set)	5,190
Total Balanced Dataset	7,785 molecules (ratio 1:2 active:inactive)
Train / Test Split	6,228 training / 1,557 testing (80/20)
Molecular Features	10-dimensional node features per atom
Fingerprint Size	2048-bit Morgan fingerprints (radius = 2)

Appendix Table C1: Dataset Summary

Appendix D: Evaluation Metrics Definitions

Metric	Definition
Accuracy	Overall fraction of correct predictions across all classes
Precision	Fraction of predicted actives that are truly active (positive predictive value)
Recall	Fraction of true actives correctly identified (sensitivity)
F1-Score	Harmonic mean of Precision and Recall; balances both types of error
AUC-ROC	Area under the ROC curve; measures discriminative ranking performance across all classification thresholds

Appendix Table D1: Evaluation Metrics Definitions

Appendix E: Tools and Libraries Used

Tool / Library	Version	Purpose
Python	3.x	Programming language
PyTorch	2.x	Deep learning framework
PyTorch Geometric	2.7.0	Graph neural network operations
RDKit	2026.3.1	Cheminformatics (SMILES parsing, fingerprints)
NumPy	2.3.5	Numerical computation
Pandas	2.3.3	Data loading and preprocessing
Scikit-learn	1.7.2	Evaluation metrics
Matplotlib / Seaborn	Latest	Visualization
Dataset Source	PubChem AID 1272365 (NCBI)	Experimental bioassay data

Appendix Table E1: Software Tools and Libraries